

TrES-3b

R-0308

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_180301 · created 2026-07-17T22:05:46+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458178.9247 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.143 +/- 0.038
Transit depth (Rp/Rs)^2	2.05 +/- 1.1 [%]
Orbital Inclination (inc)	83.15 +/- 1.9
Airmass coefficient 1 (a1)	0.984 +/- 0.017
Airmass coefficient 2 (a2)	0.0099 +/- 0.0051
Scatter in the residuals of the lightcurve fit is	1.73 %
Best Comparison Star	#3 - [70, 201]
Optimal Method	PSF photometry
Transit Duration (day)	0.06 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.143 ± 0.038 vs 0.16309 (deviation 0.46σ)
Duration fit vs published	0.06 d vs 0.0567 d (deviation 0.17σ)
Mid-time O–C	2.3 min
Depth SNR	3.3
Residual scatter	1.73 %

Light curve

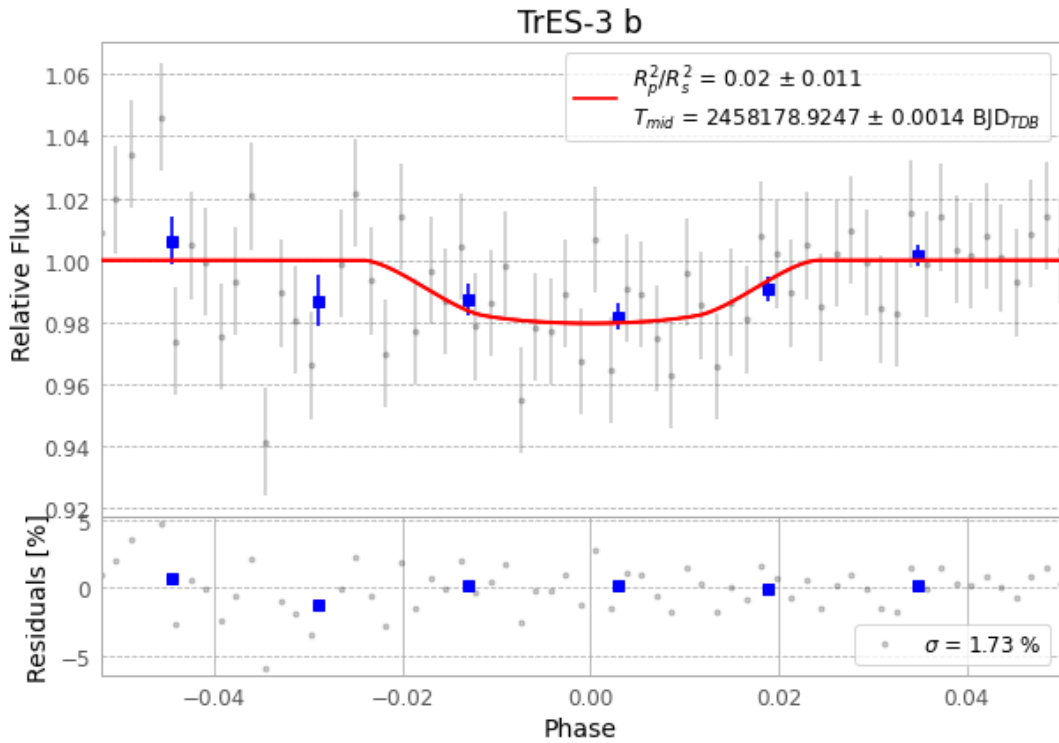


Figure 1 — FinalLightCurve_TrES-3 b_2018-03-01.png (SHA-256 aa6965ed8feb65d3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [409, 155], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c5f8e9a1d0763b29...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 055cf2e

Data provenance

65 FITS frames (43 MB), TRES-3180301083312.FITS → TRES-3180301114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-180301.log (2f8f7faa8ead...), FinalLightCurve_TrES-3 b_2018-03-01.png (aa6965ed8feb...), FinalLightCurve_TrES-3 b_2018-03-01.csv (4f196a160aa6...)

Compute resources

Wall time 130.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 22:05Z.